BADG 2: A Perfect Game

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Abstract

An earlier (b. a. d. g.) paper (http://fettermania.com/math/bdice.pdf) focused on a near-optimal strategy for maximizing expected value in gameplay. This paper focuses on a different question: what is the probability of playing a perfect game? Using standard recursive calculation, we calculate the probabilities with the out-of-box game setup. Then we venture into a related: What's the expected win probability with n standard dice of side count s as $n \to \infty$? We give poor bounds and prove convergence.

1 Introduction and the Commercial Game

Outside of ideological patron Larry Waldman, the publication of my original b. a. d. g. paper (http://fettermania.com/math/bdice.pdf) received little human fanfare, as expected. However, robots.txt somehow picked up on this and brought this to the attention of Samuel Cox, creator of the game. After dispensing with mild praise about the paper and strong confusion about my interest in it, Mr. Cox suggested another problem.

Instead of "how does one play the game with high/maximum expected value?" (addressed in the first paper), Mr. Cox wondered how often a *perfect game* (one in which the best possible score is achieved at the end) occur? I took this on and a day or so later, the solution to this particular question surfaced readily. Naturally unsatisfied by instances, I sought a more general solution to a subproblem I found. Nine months later, just as I was beginning to enjoy my futility and disinterest in this neglected offspring, the brood emerged.

So we'll briefly discuss the game, the assumption, the solution method for the commercially offered game, the numbers before diving in to the second quest and surprising discoveries found along the way.

1.1 The Game

With no access to the game, we search the web. From what I can only assume is/was the front page of boardgamegeek.com, b.a.d.g. has these drunkenly paraphrased rules (sic):

1. *Roll all the dice*
2. *Determine if you have bitches a "bitch" is the highest number on any dice usually this is the six but there are a few exceptions the 8 and 10 and 12-sided die as well as the custom*
3. *Remove dice you want to set aside your highest dice to get the best past possible score. You must set aside at least one die per roll you can set aside as many dice as you want.*
4. *Reroll the remaining dice and keep setting them aside until you're out of dice.*
5. *Score add up all the dice The person with the highest score wins.*

I'll take it from here, TrevDogg. To formalize this more clearly, a solitaire-style game of b. a. d. g. follows this pattern:

1. Begin with 12 standard dice of six sides, along with one eight-, one ten-, and one twelve-sided die.
2. On each of your turns, roll all of the dice.
3. If you set any dice from this roll aside, add their face total to your overall score.
4. You must pick up at least one die each turn.

The question *"how likely am I to get a perfect score?"* therefore relies on the *strategy* one employs in picking up dice.

1.2 The Assumption

Proposition 1.1 (The Smoke 'em if you got 'em strategy)**.** *: The best strategy to achieve a perfect game is to always pick up as many dice with maximum face value as the turn allows.*

Upon reflection, this proposition is:

1. Simple
2. Obvious
3. The strategy any reasonable person would employ

4. Assumed throughout the rest of the paper
5. **Mathematically false**

Why mathematically false? The second quest offers a few surprises. See **The Smoking Gun** section for details. However, for the discretely sized dice of 6, 8, 10, and 12 sides, we'll assume having more points in the bank and less dice to roll is better than the inverse; reasonably, if you set aside a die with a max roll, that die will *again* have to hit a max roll to achieve your perfect game. Why not just bank it?

So, we assume it in calculating the expected value of any state reachable from the standard b. a. d. g. starting configuration in the next section.

1.3 The Commercial Game Solution Method

These definitions will help us write a program to get our solution for the commercial game:

- The "state" $\vec{S} := [a, b, c, d]$ is how many dice of size 6, 8, 10, and 12 remain, respectively. We start at $\vec{S} = [12, 1, 1, 1]$. A perfect game ends at $\vec{S} = \vec{0}$. A failure to roll any max roll ends the game in an unlisted 'losing' state $\varnothing$.
- The "probability vector" $\vec{p} := [1/6, 1/8, 1/10, 1/12]$ is the chance of getting a max roll from each of the dice sizes, again here assumed to be 6, 8, 10, 12.
- With a win defined as value 1 and a loss as value 0, the "expected value" of being in state $\vec{s}$ is defined as $F(\vec{s})$. Since there is only a single, deterministic strategy, the expected value is then equivalent to the probability of ending up in winning state $\vec{0}$ from state $\vec{s}$.
- Therefore, $F(\vec{0}) = F([0, 0, 0, 0]) = 1$, and $F(\varnothing) = 0$.

First, let's limit ourselves to a game (or state with only $d - sided$ dice, collapsing state vector $\vec{s}$ to a single integer r (r dice of size d left).

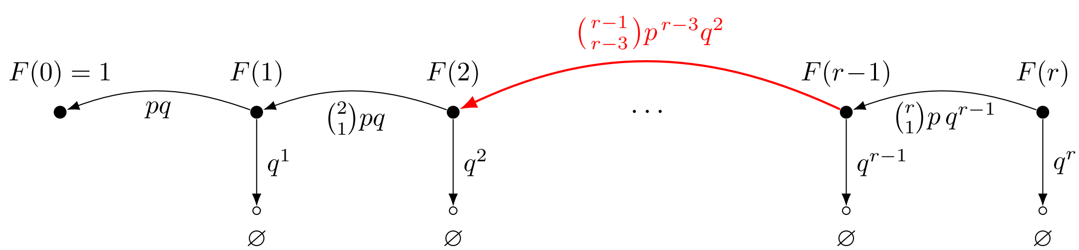

Figure 1: Transition graph of the dice game showing expected values $F(t)$ and transition probabilities.

Lemma 1.2 (EV of r d-sided dice)**.** *: The expected chance of a completing perfect game from a state of r standard dice of d sides each, given $p := \frac{1}{d}$ and $q := 1 - p$, is $F(k) = \Sigma_{k=1}^{r} \binom{r}{k} p^k q^{r-k} F(r-k)$, with $F(0)$ defined as 1.*

Proof: This a clear application of the binomial theorem. The chance of transitioning to state $F(r-k)$ from $F(r)$ is the chance of rolling exactly k max rolls and $r-k$ non-max rolls among your d-sided dice, or $P_{single}(r, k, p) := \binom{r}{k} p^k q^{r-k}$. The expected value of being in this new state is defined as $F(r-k)$. The value of this state transition possibility is their product, and the total expected value of being in state r is the sum of the values of all state transition values. Note that $k = 0$ implies a roll with no maxes, and thus a loss (value 0).

Extending this to dice of different sizes is clear:

Lemma 1.3 (State Transition lemma)**.** *The chance of transitioning from state $\vec{s_1} = s_{11}, s_{12} \ldots s_{1z}$ to state $\vec{s_2} = s_{21}, s_{22} \ldots s_{2z}$ with probability vector $\vec{p} = p_1, p_2 \ldots p_z$ is the product of the individual probabilities: $P_{transition}(\vec{s_1}, \vec{s_2}, \vec{p}) := \Pi_{i=0}^{z} P_{single}(\vec{s_1}_i, \vec{s_2}_i, \vec{p}_i)$.*

Proof: Since the outcome of rolling dice of d sides are all indepedent of each other, $P_{transition}$ can be expressed as the product of P_{single} quantiites.

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This is solvable using standard recursive techniques.

Algorithm 1 Dynamic program for $F(\vec{s})$

$\vec{S} \leftarrow [12, 1, 1, 1]$
$\vec{p} \leftarrow \left[\frac{1}{6}, \frac{1}{8}, \frac{1}{10}, \frac{1}{12}\right]$
$P_{\text{single}}(n, k, p) \leftarrow \binom{n}{k} p^k (1-p)^{n-k}$
$P_{\text{transition}}(\vec{s_1}, \vec{s_2}, \vec{p}) \leftarrow \prod_{i=0}^{\dim(\vec{s})} P_{\text{single}}(\vec{s_1}_i, \vec{s_2}_i, \vec{p}_i)$
$F(\vec{0}) \leftarrow 1$
$\text{substates}(\vec{s}) \leftarrow \prod_{i=0}^{\dim(\vec{s})} [0, s_i]$
$\text{states} \leftarrow \text{topological\_sort}\Big(\text{substates}(\vec{S}) \setminus \{[0,0,0,0]\}\Big)$
for $\vec{s} \in$ states **do**
    $F(\vec{s}) \leftarrow \sum_{\vec{t} \in (\text{substates}(\vec{s}) \setminus \{\vec{s}\})} P_{\text{transition}}(\vec{s}, \vec{t}) \cdot F(\vec{t})$
end for

EV with $(d_{10}, d_{12}) = (0, 0)$

$d_6 \backslash d_8$	0	1
0	1	0.125
1	0.1667	0.05642
2	0.07407	0.03244
3	0.04192	0.02206
4	0.02809	0.01679
5	0.02107	0.01378
6	0.01708	0.01195
7	0.01463	0.01076
8	0.01303	0.009966
9	0.01195	0.009423
10	0.01119	0.009044
11	0.01065	0.008779
12	0.01025	0.008591

EV with $(d_{10}, d_{12}) = (1, 0)$

$d_6 \backslash d_8$	0	1
0	0.1	0.03469
1	0.04556	0.02046
2	0.02644	0.01425
3	0.01815	0.01109
4	0.01392	0.009299
5	0.01152	0.008218
6	0.01006	0.007537
7	0.009124	0.0071
8	0.008506	0.006818
9	0.00809	0.006638
10	0.007809	0.006529
11	0.007618	0.006468
12	0.007491	0.006441

EV with $(d_{10}, d_{12}) = (0, 1)$

$d_6 \backslash d_8$	0	1
0	0.08333	0.02908
1	0.03819	0.01726
2	0.02231	0.01209
3	0.0154	0.009464
4	0.01188	0.00798
5	0.009888	0.007088
6	0.008677	0.006532
7	0.007907	0.006181
8	0.007404	0.00596
9	0.007071	0.005825
10	0.006851	0.00575
11	0.006707	0.005715
12	0.006618	0.005709

EV with $(d_{10}, d_{12}) = (1, 1)$

$d_6 \backslash d_8$	0	1
0	0.02347	0.01087
1	0.01406	0.007809
2	0.009946	0.006252
3	0.007849	0.005385
4	0.006672	0.004879
5	0.005971	0.00458
6	0.005542	0.004408
7	0.005278	0.004318
8	0.00512	0.004283
9	0.005033	0.004284
10	0.004994	0.004312
11	0.004988	0.004356
12	0.005005	0.004413

Lemma 1.4 (Easy Cases Lemma)**.** *If $p = 0$, $F(r) = 0$. If $p = 1, F(r) = 1$. If $s = 2$, for any $r > 0, F(r) = \frac{1}{2}$.*

Proof: Though we won't be flipping infinite ($p = 0$) or 1-sided ($p = 1$), dice, these algebraic identities are obvious (can't succeed, can't fail, respectively) and will be useful later.

Base case for s=2: $F(1) = \frac{1}{2}$ since a perfect game occurs if and only if the coin flips 2 instead of 1.

Inductive case for s=2: Assuming $F(k) = \frac{1}{2}$ for all $0 < k < r$, and $F(0) = 1$:

$$\Sigma_{k=1}^{r-1}\binom{r}{k}p^k q^{r-k}F(r-k) + \binom{r}{r}p^r q^0 F(0) = \tag{1}$$

$$(2^r - 2)\frac{1}{2^r}(\frac{1}{2}) + \frac{1}{2^r}2(\frac{1}{2}) = \tag{2}$$

$$(2^r)\frac{1}{2^r}(\frac{1}{2}) = \frac{1}{2} \tag{3}$$

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- Initial Problem Statement: Perfect games
- Solution for BADG commercial version
- One die type, $n \to \infty$
- Easy Cases
- Lazy Upper bound

- Lazy Lower bound (Pochhammer)
- Extremely Lazy Lower bound (closed form)
- Wrong attempt: $F(t) < F(t-1)$?
- F(n) exhibits provably asymptotic behavior

Theorem 1.5. *For an unbounded set of $r > 0$ standard s-sided dice, $s \geq 2$, the probability $F(r)$ of getting a perfect game has a lower bound of* $\exp(-\frac{q}{(1-q)^2})$*, if we define $q := 1 - \frac{1}{s}$.*

Proof:

- First, while at stage k, where k dice remain, the probabilty $P_s(k)$ of continuing to some stage $j < k$ is clearly $1 - q^k$.
- The probability $F(k)$ of passing all such stages is greater than or equal to $P^* = \Pi_{i=1}^{k} P_s(k)$, since a successful traversal may involve skipping one or more stages, and P^* assumes we no stages.
- Therefore, $F(k) \geq \Pi_{i=1}^{k} P_s(k) = \Pi_{i=1}^{k}(1 - q^k)$.
- This means $\log(F(k)) \geq \Sigma_{i=1}^{k} \log(P_s(k)) = \Sigma_{i=1}^{k} \log(1 - q^k)$.

This $F(k)$ is apparently defined as the **q-Pochhammer symbol** $(q;q)_\infty = \Pi_{i=1}^{k}(1 - q^k)$, with known convergence properties. I'll instead attempt to tackle convergence without the aid of these theorems.

Often, $\log(1-z)$ is approximated by $-z$ near zero, but we are most concerned with z near one, since $1 - z$ approaches 0, and $\log(1-z)$ explodes. Therefore, let's look at the expansion of $\log(1-z)$:

- Identity 1: $f(z) = -\frac{1}{1-z} = -1 - z - z^2 - \ldots$
- Identity 2: $\int f(z) = \log(1-z) = -z - \frac{z^2}{2} - \frac{z^3}{3} - \ldots$

So when summing $\Sigma_{i=1}^{k} \log(1-q^k)$, we can instead sum

$$\log(F(k)) = \log(1-q^1) + \log(1-q^2) + \log(1-q^3) + \dots \tag{4}$$

$$\log(F(k)) = (-q - \frac{q^2}{2} - \frac{q^3}{3} + \dots) + (-q^2 - \frac{q^4}{2} - \frac{q^6}{3} + \dots) + (-q^3 - \frac{q^6}{2} - \frac{q^9}{3} + \dots) + \dots \tag{5}$$

$$> (-q - q^2 - q^3 + \dots) + (-q^2 - q^4 - q^6 + \dots) + (-q^3 - q^6 - q^9 + \dots) + \dots \tag{6}$$

$$= \frac{-q}{1-q} + \frac{-q^2}{1-q^2} + \frac{-q^3}{1-q^3} + \dots \tag{7}$$

$$> \frac{-q}{1-q} + \frac{-q^2}{1-q} + \frac{-q^3}{1-q} + \dots \tag{8}$$

$$= \frac{1}{1-q}[-q - q^2 - q^3 - \dots] \tag{9}$$

$$= \frac{-q}{(1-q)^2} \tag{10}$$

- (2) follows from (1) by substitution using Identity1 above.
- (3) follows from (2) since the denominators of (2) are larger.
- (4) follows from (3) when dividing each sequence by $-q^k$ and applying identity 2.
- (5) follows from (4) since the denominators of (4) are larger.
- (6) and (7) follow from (5) by dividing out $\frac{1}{1-q}$ and applying Identity 1.

Since $\log(F(k)) > \frac{-q}{(1-q)^2}$, $F(k) > \exp(\frac{-q}{(1-q)^2)})$.

s	p	q	bound
2	$\frac{1}{2}$	$\frac{1}{2}$	$\exp(-2) \approx .135$
3	$\frac{1}{3}$	$\frac{2}{3}$	$\exp(-6) \approx .00248$
4	$\frac{1}{4}$	$\frac{3}{4}$	$\exp(-12) \approx 6.14 \times 10^{-6}$
10	$\frac{1}{10}$	$\frac{9}{10}$	$\exp(-90) \approx 8.19 \times 10^{-40}$

Table 1: Very loose lower bounds on $F(k = \infty)$

This is a thoroughly unsatisfying lower bound, as we know $F(2) = \frac{1}{2}$.

Definition 1.6 ($(q;q)_\infty$: The q-pochhammer symbol)**.** *The q-pochhammer symbol* $(q;q)_\infty = \Pi_{k=1}^{\infty}(1-q^k)$ *is a known convergent quantity.*

q	$(q;q)_\infty$	$\exp\left(-\frac{q}{(1-q)^2}\right)$
1/2	2.8879×10^{-1}	1.3534×10^{-1}
2/3	6.9272×10^{-2}	2.4788×10^{-3}
3/4	1.5545×10^{-2}	6.1442×10^{-6}
4/5	3.3680×10^{-3}	2.0612×10^{-9}
5/6	7.1400×10^{-4}	9.3576×10^{-14}
6/7	1.4913×10^{-4}	5.7495×10^{-19}
7/8	3.0815×10^{-5}	4.7809×10^{-25}
8/9	6.3155×10^{-6}	5.3802×10^{-32}
9/10	1.2861×10^{-6}	8.1940×10^{-40}

However, this does provide a positive lower bound, and brings up this interesting fact: **Though increasing the number of sides of the dice can drive F(k) to zero, increasing k to ∞ will never set F(k) to zero**.

If we can couple this with the proof of the next theorem, we can state that $F(k)$ has an asymptotic limit as $k \to \infty$ given some q.

Calculating by hand:

- $F(0) = 1$
- $F(1) = p$
- $F(2) = p^2(1+2q) = p^2(3-2p)$
- $F(3) = p^3(13 - 27p + 21p^2 - 6p^3)$
- $F(4) = p^4(75 - 316p + 606p^2 - 664p^3 + 432p^4 - 156p^5 + 24p^6)$
- $F(1) - F(0) = p - 1 = -(1-p)$
- $F(2) - F(1) = p^2(3-2p) - p = (-1)p(1-2p)(1-p)$
- $F(3) - F(2) = (-1)3p^2(1-p)^3(1-2p)$
- $F(4) - F(3) = -p^3(1-2p)(p-1)^2(12p^4 - 48p^3 + 72p^2 - 50p + 13)$

Theorem 1.7 (Incorrect Blind Alley: $F(n) < F(n-1)$)**.** *If* $p < \frac{1}{2}, F(n-1) > F(n)$ *for* $n > 1$.

- TODO: Show calculation of various functions directly.
- Show calculation of $p = \frac{1}{2} - \epsilon, q = \frac{1}{2} - \epsilon$ with $p = 0.495$.

- Show interesting observations: growth of polynomial
- $R(t)$ is starting at position t, throwing $t-1$ dice.
- $F(t) = p(1+q^{t-1})F(t-1) + qR(t) = pF(t-1) + pq^{t-1}F(t-1) + qR(t)$
- If $qR(t) < qF(t-1) - pq^{t-1}F(t-1)$, same as $F(t-1) - R(t) > pq^{t-2}F(t-1)$, then $F(t) < F(t-1)$.
- In the special case $q = p = \frac{1}{2}$, then $F(t-1) - R(t) = p^{t-1}(1) - p^{t-1} \cdot p$ + other zero factors $= p^t = pq^{t-2} \cdot \frac{1}{2}$.
- So we're satisfied since $F(t) = \frac{1}{2}$ for $t \geq 1$.
- If $p \leq \frac{1}{3}$, $F(t-1) - R(t) = [p^{t-1}F(0) - p^{t-1}F(1)$ + other positive factors $] = p^{t-1}q + \epsilon$.

$$t = 3, F(3) - R(4) > pq^{4-1}F(4-1) \tag{11}$$

$$p^3F(0) - p^3F(1) \tag{12}$$

$$\binom{3}{2}p^2qF(1) - \binom{3}{2}p^2qF(2) \tag{13}$$

$$\binom{3}{1}p^2qF(2) - \binom{3}{2}p^2qF(3) \tag{14}$$

$$\tag{15}$$